

Syed Muhammad Taha Imam

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PROFESSIONAL SUMMARY

AI/ML Engineer specializing in building and deploying production AI systems on Google Cloud Platform, with hands-on experience across Vertex AI, the Gemini API, and RAG architectures. Delivered conversational AI products used by hundreds of daily users, including safety-critical deployments with strict content-moderation guardrails for regulated, multi-audience use cases. Strong in end-to-end MLOps: containerized deployment (Docker, Cloud Run, GitHub Actions CI/CD), vector search, and multi-model orchestration for cost and latency optimization. Research background fine-tuning foundation models, with healthcare-domain experience evaluating the factual accuracy of clinical AI outputs.

EDUCATION

2022 – 2026 BSCS at **National University of Sciences & Technology** (cGPA: 3.5/4.0)

Relevant Coursework: Linear Algebra, Probability & Stats, Deep Learning, Machine Learning

Ranked 20th/127, Executive @ NUST Science Society, Team Lead @ GDSC

2020 – 2022 A'Levels at **Cedar College** (GPA: 4.33/4.0, Grade: 3A*)

INDUSTRY EXPERIENCE

AI Engineer, Brainbox Automations Dec 2025 – Present

- Deployed a safety-guardrailed AI coaching backend on GCP Cloud Run (Dockerized FastAPI, Vertex AI embeddings) serving 300+ minor and adult athletes, engineering content-moderation guardrails to flag offensive and sexual content for compliant, multi-audience deployment; optimized its GitHub Actions CI/CD pipeline to cut deployment time from ~ 2 minutes to 10 seconds, retaining an SSH-based manual script as an emergency fallback.
- Engineered SQL-aware agents for a Supabase-hosted RAG chatbot across 50+ tables, achieving superior query accuracy and reducing response time by 50%.
- Partnered with Earlibird, an Australian automation firm, to build a customer-facing chat and VAPI voice AI system for brand queries, appointment booking, and direct product search, handling 200+ daily interactions (80% full AI resolution, 21% booking conversion, 32% escalation) while processing 5,000+ AI-handled inbound and outbound calls and saving clients 500+ manual hours.
- Utilized AWS Lambda functions with background task queues to summarize hour-plus sales-call transcripts via the Gemini API using a parallel map-reduce strategy to work around context-length limits and timeouts.

AI Fullstack Engineer, Epistemy Sep 2025 – Dec 2025

- Led end-to-end development of an AI tutoring platform, architecting a Nest.js backend (20+ endpoints, Supabase) and a Next.js frontend.
- Architected an event-driven task queue (Redis/BullMQ) to orchestrate multi-agent workflows, ensuring fault tolerance across distributed analysis pipelines.
- Enforced strict SWE standards with CI pipelines and pre-commit hooks, achieving and maintaining 90%+ unit test coverage.

Machine Learning Intern, CogniMind AI Feb 2025 – Apr 2025

- Automated MLOps by deploying dockerized Apache Airflow and designing 5+ DAGs, reducing manual intervention by 60%; built CI/CD pipelines (Docker, GitHub Actions) cutting deployment time by 30%.
- Improved model performance by enhancing VLM extraction accuracy by 20% (prompt engineering) and optimizing retrieval/inference pipelines by 10% (quantization, HNSW).

Machine Learning Intern, RapidsAI Sep 2024 – Dec 2024

- Developed and integrated a RAG-based chatbot (Streamlit, FastAPI) with contextual session management, using Chain-of-Thought prompting to reduce errors by 50% and a multi-model, complexity-based query router to cut OpenAI API costs 50%.

RESEARCH EXPERIENCE

Research Intern, IPT Lab, NUST	Jun 2026 – Present
– Fine-tuning pre-trained wireless/RF foundation models for downstream sensing tasks, including human activity recognition (HAR), 16-way beam prediction, etc., adapting large-scale foundation models to resource-constrained sensing applications. – Applying parameter-efficient fine-tuning (PEFT) techniques to adapt models under limited compute and labeled data, evaluating performance across multiple downstream sensing benchmarks.	
Undergraduate Research Intern, MachVis	Oct 2025 – Dec 2025
– Proposed a novel LLM-based factual verification framework to evaluate the clinical accuracy of pathology reports, moving beyond linguistic metrics (BLEU/ROUGE). – Curated a “Challenge Dataset” of 100+ gigapixel whole-slide images (WSIs) featuring real-world artifacts and stain variations to assess model robustness.	

PUBLICATIONS

[1] Tayyab Raza, **Syed Muhammad Taha Imam**, et al. *LiteDoc: Distilling Large Document Models into Efficient Task-Specific Encoders*. Accepted: International Conference on Document Analysis and Recognition (ICDAR). Aug. 2026.

[2] Nuzhat Akram, **Syed Muhammad Taha Imam**, et al. *Accuracy of Craniometric Features in Gender Estimation Using Machine Learning Algorithms on University of Tennessee (UT) and Howells Datasets*. Manuscript in preparation. 2026.

SKILLS

Cloud & MLOps	Google Cloud Platform (Vertex AI, Cloud Run); AWS (Lambda, EC2); Docker; GitHub Actions CI/CD; Apache Airflow
LLMs & Agentic AI	Gemini API; OpenAI/Groq APIs; RAG; Prompt Engineering; LangChain; Multi-Agent Workflows; vLLM/Triton
Core ML	Python; PyTorch; TensorFlow; Hugging Face; Model Fine-Tuning (PEFT); Quantization
Backend & Automation	FastAPI; Nest.js; Redis/BullMQ; Supabase; VAPI; Make.com; n8n

PROJECTS

create-rag-app	Syed-MuhammadTaha/create-rag-app
– Built a Python CLI (Typer) to generate production-ready, Dockerized RAG apps with 8+ integrations (OpenAI, Qdrant) and automated full-stack scaffolding (FastAPI, Streamlit). – Designed a modular architecture using abstract base classes and 13 Jinja2 templates to support 3 distinct retrieval strategies. – Tools Used: Python, Typer, Docker, FastAPI, Streamlit, Qdrant, LangChain, Jinja2.	
Sable Prospector	Syed-MuhammadTaha/baremetal-prospector-agent
– Engineered a framework-free autonomous SDR agent utilizing a raw Python ReAct loop with a syntax-directed parser and auto-healing mechanisms to prevent tool hallucinations, backed by a three-tier memory system (sliding-window summaries, a bare-metal vector database, and a JSON long-term store). – Tools Used: Python, Jina Embeddings, OpenAI/Groq APIs, Streamlit, uv.	

ACHIEVEMENTS

- **5x Dean’s List recipient:** Awarded for academic excellence in a particular semester.
- **Hackathon Winner & Internship Recipient:** Secured a paid internship by winning a week-long hackathon against 70+ participants (Epistemy).
- **Finalist, VisionRD Hackathon:** Developed an action recognition pipeline with a custom backbone.
- **Technical Writer at Medium (@tahaML):** Authored multiple articles on GenAI and Deep Learning, attracting 100+ views.

LANGUAGES & CERTIFICATIONS

Languages	English (Advanced), Urdu (Native), German (Beginner)
Certifications	Deep Learning Specialization (deeplearning.ai)